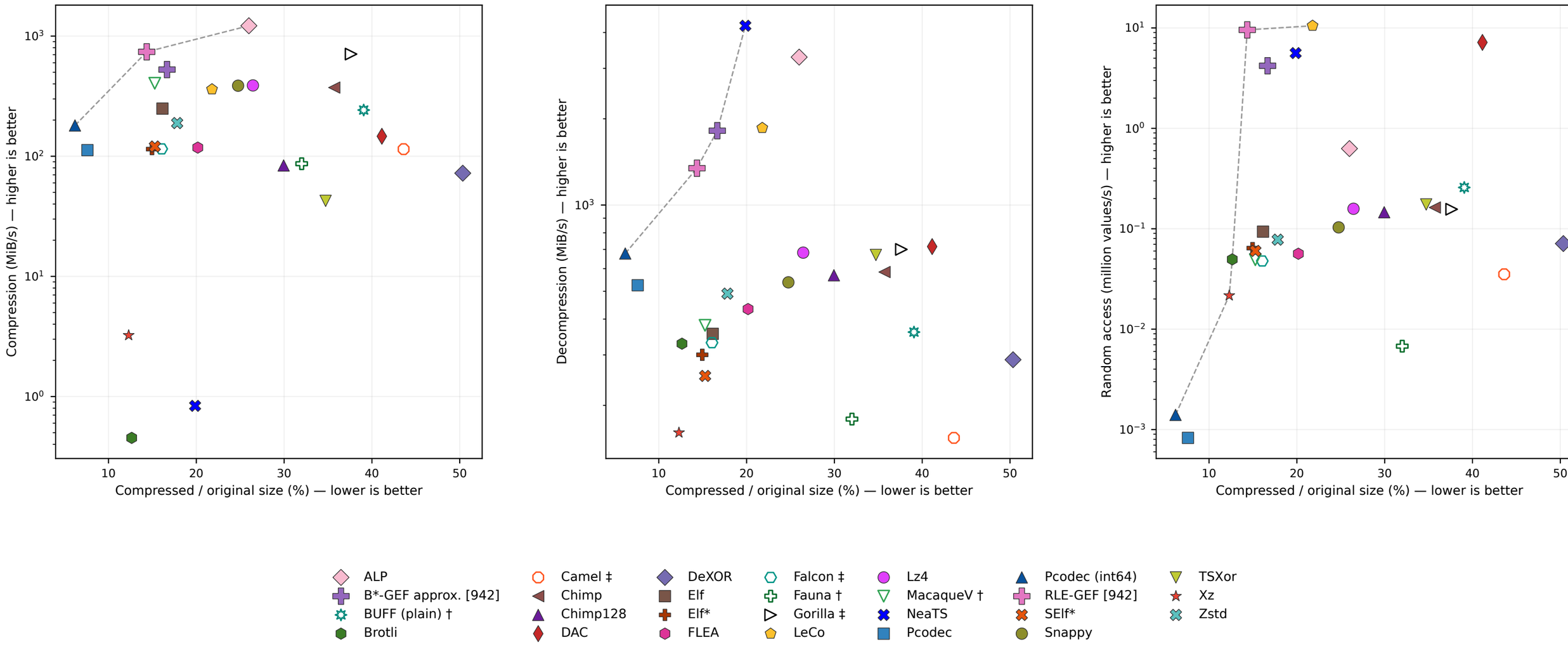
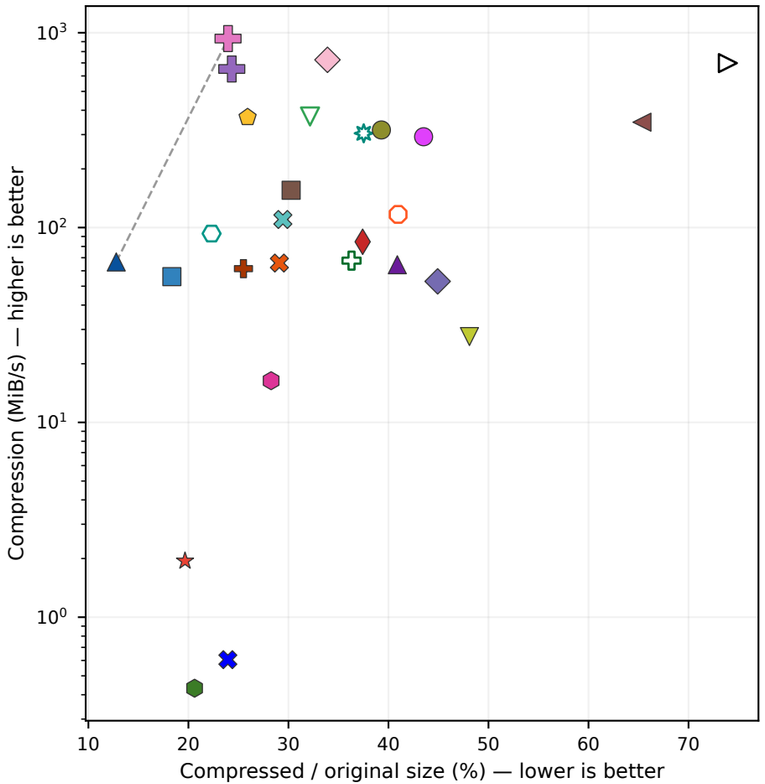
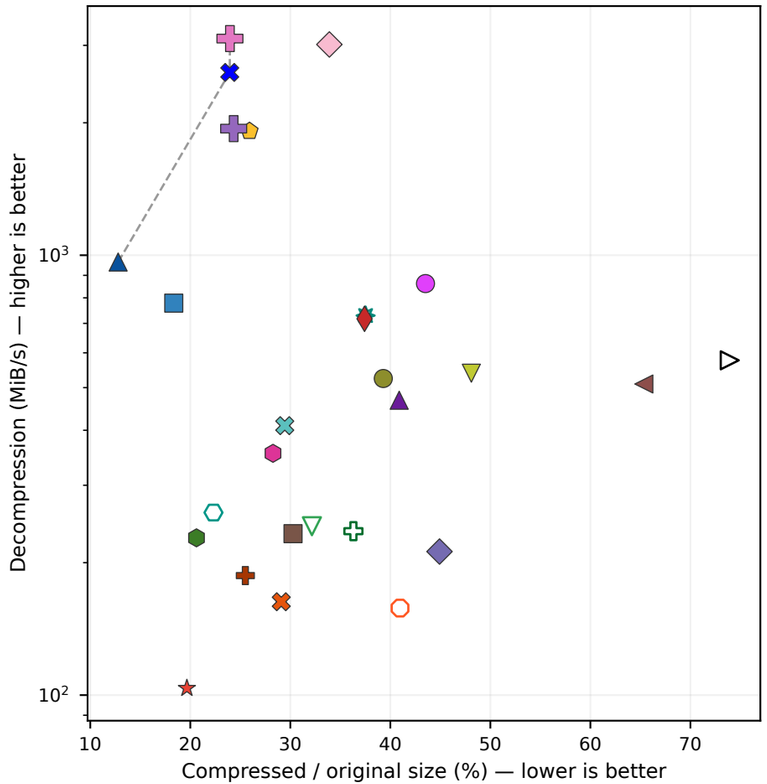
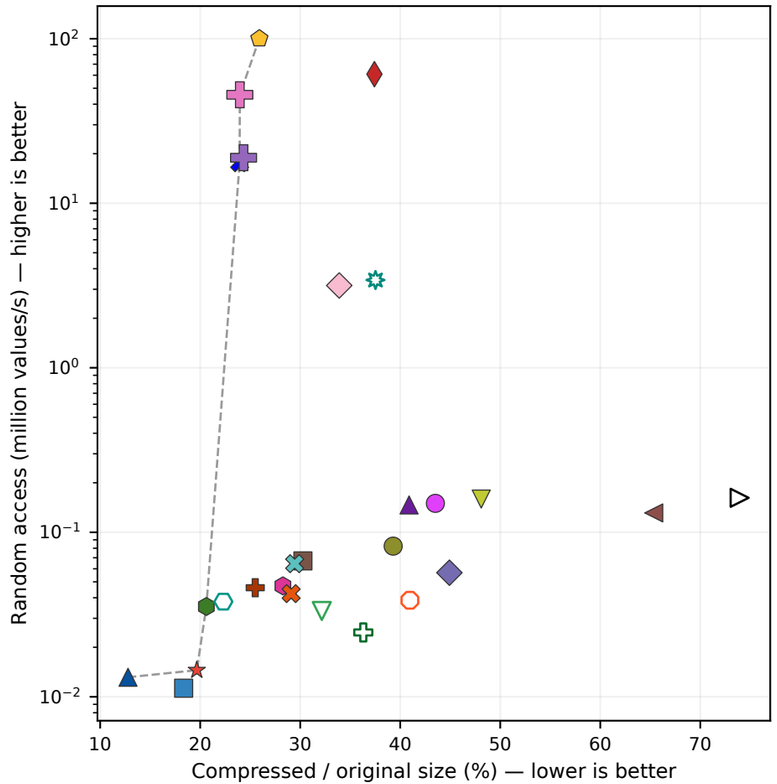

AP • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

BM • native GEF 942d3c4 (uniform K=32,000) vs competitors

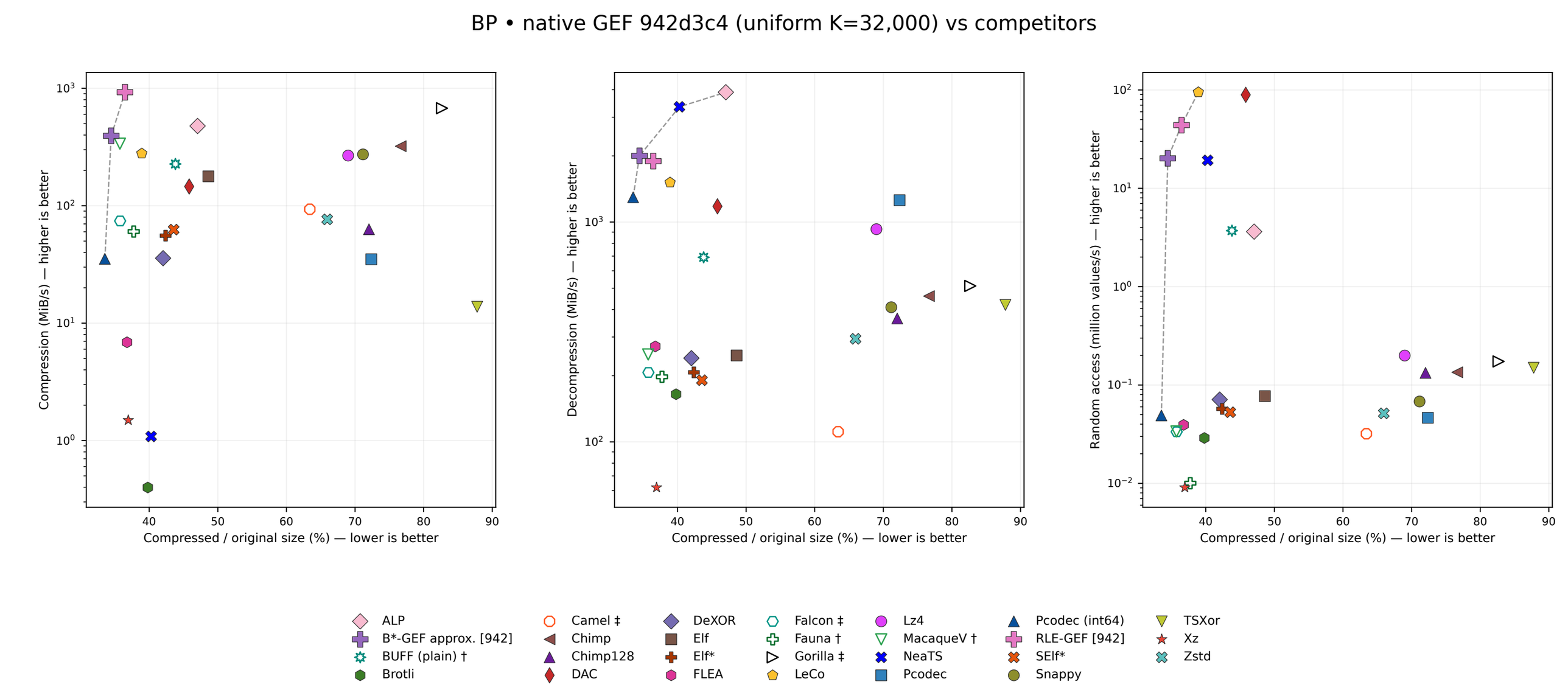




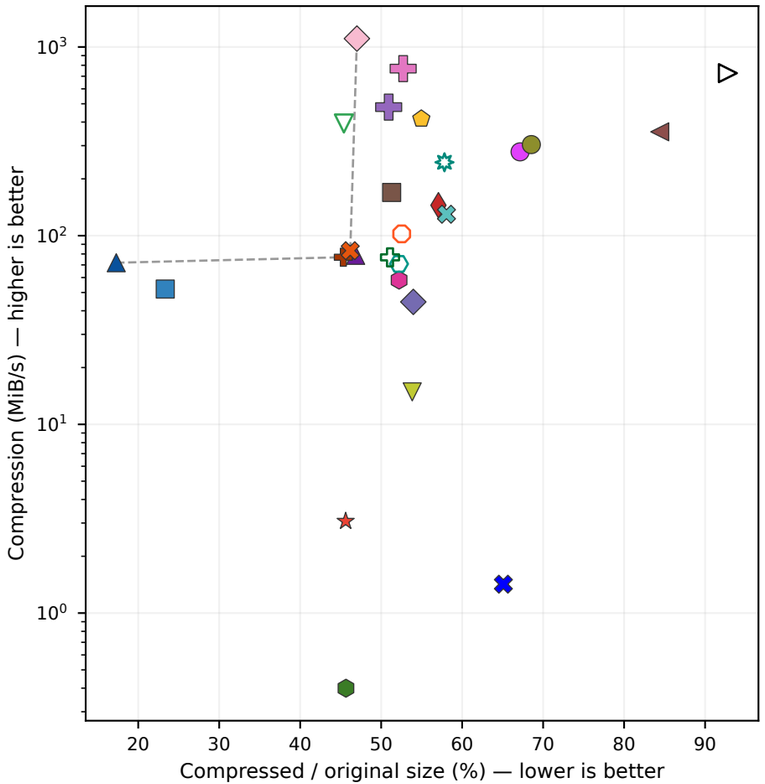


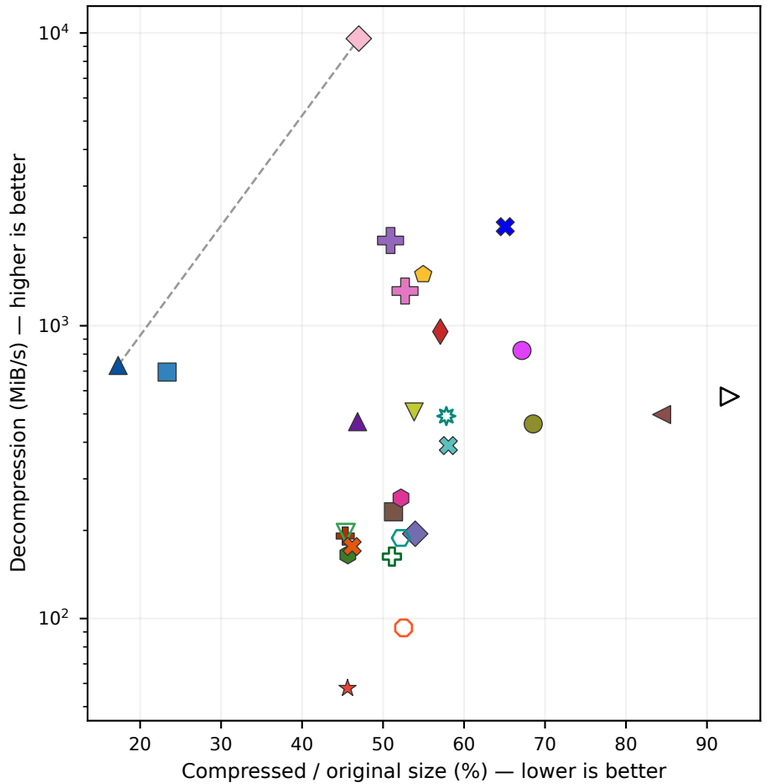
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

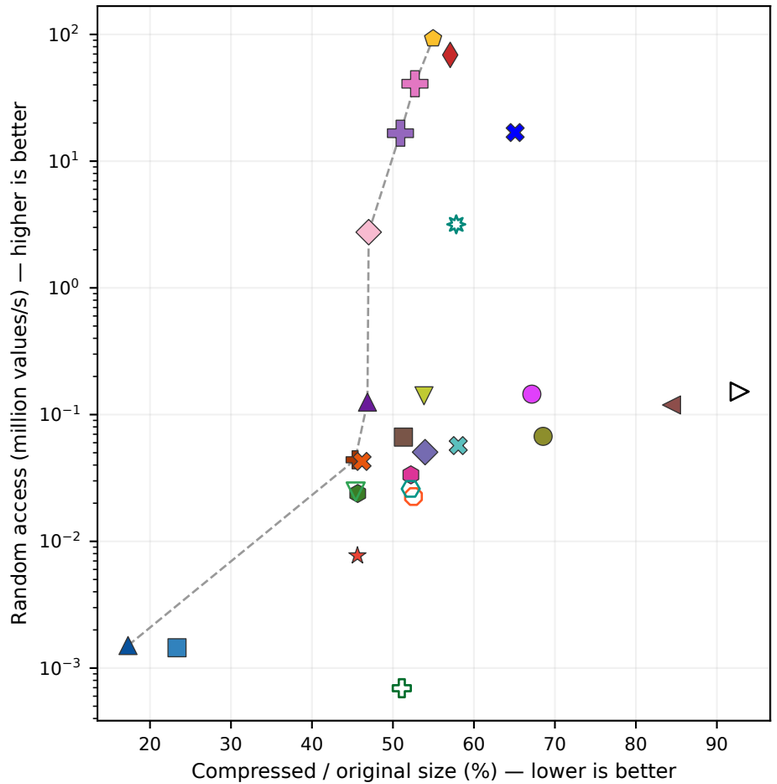
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.


BP • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

BT • native GEF 942d3c4 (uniform K=32,000) vs competitors

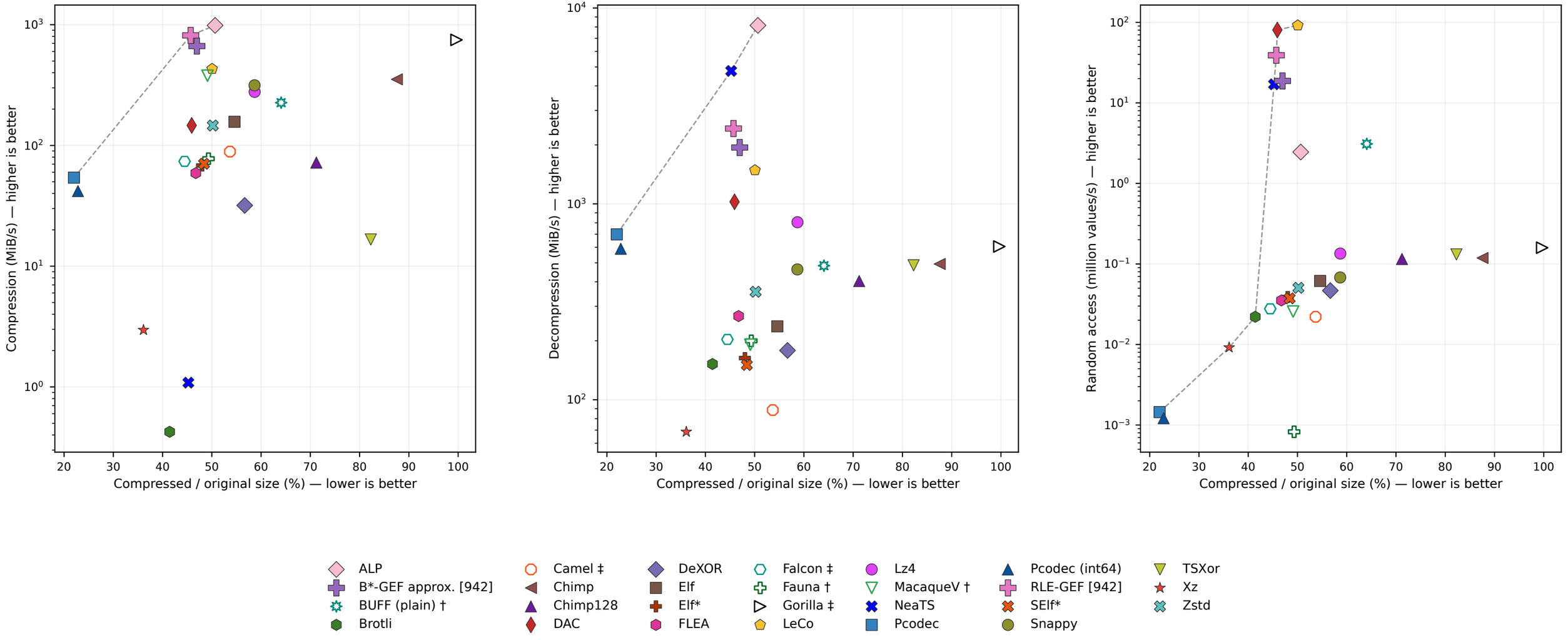




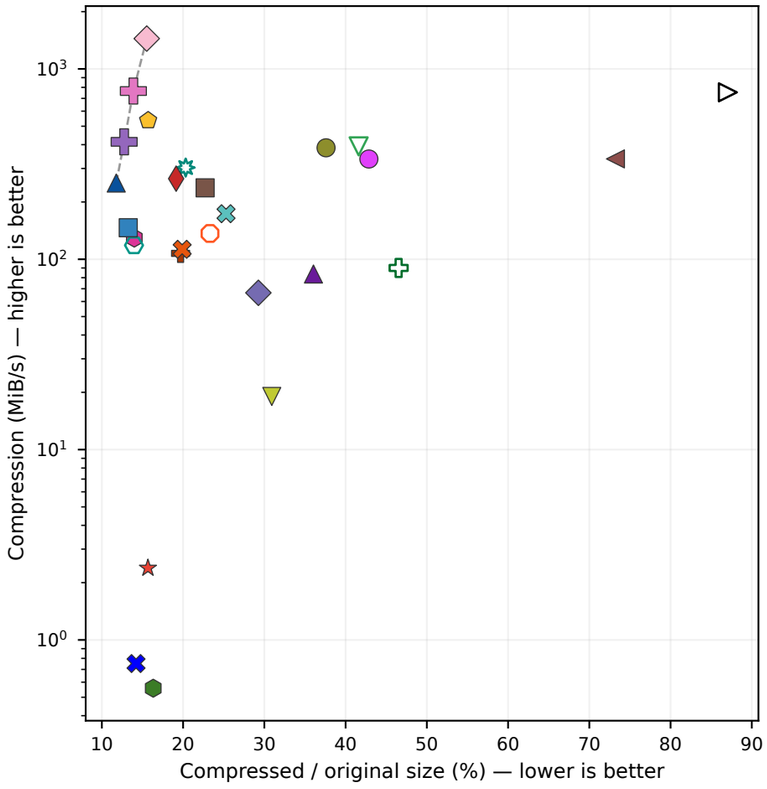


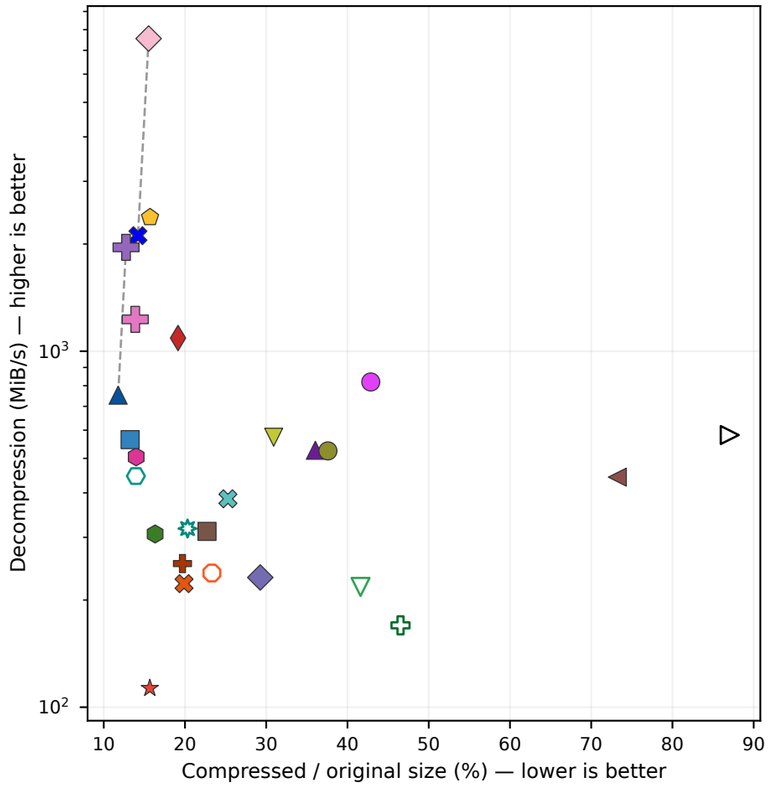
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

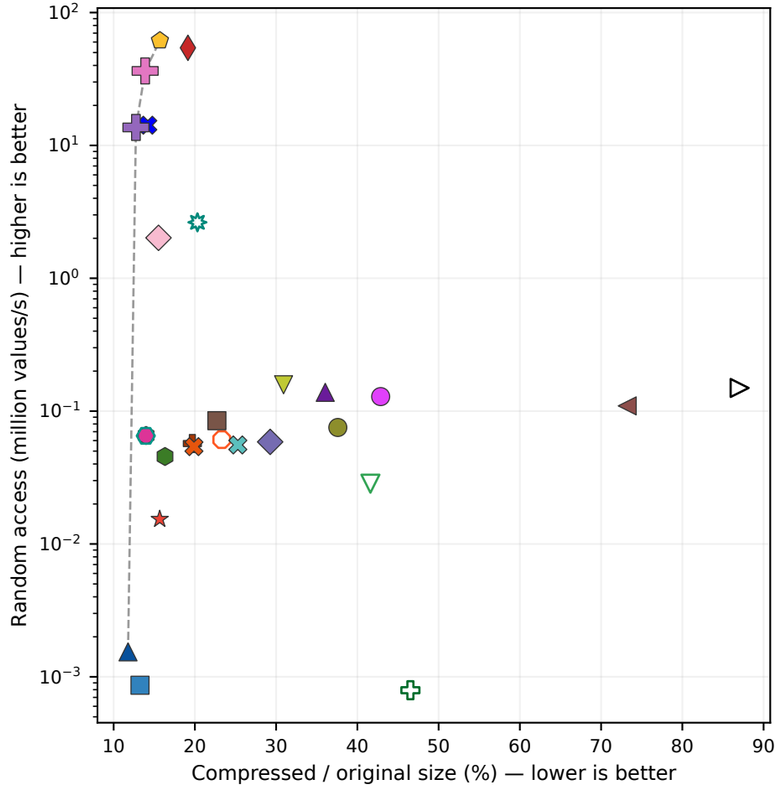
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.


BW • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

CT • native GEF 942d3c4 (uniform K=32,000) vs competitors



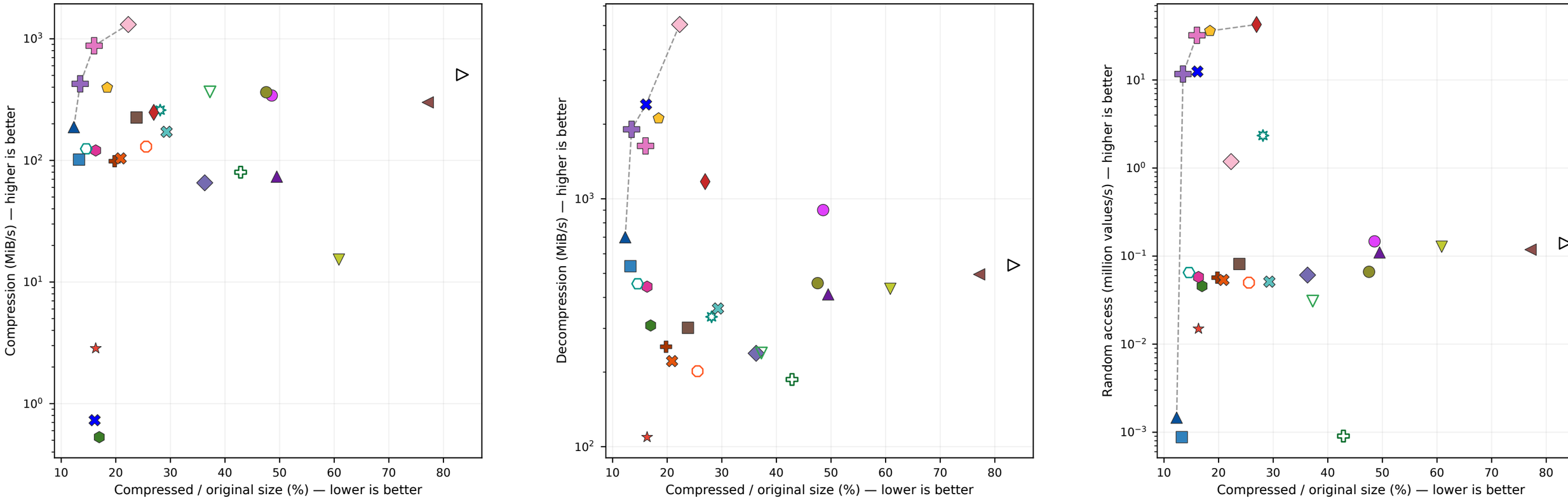




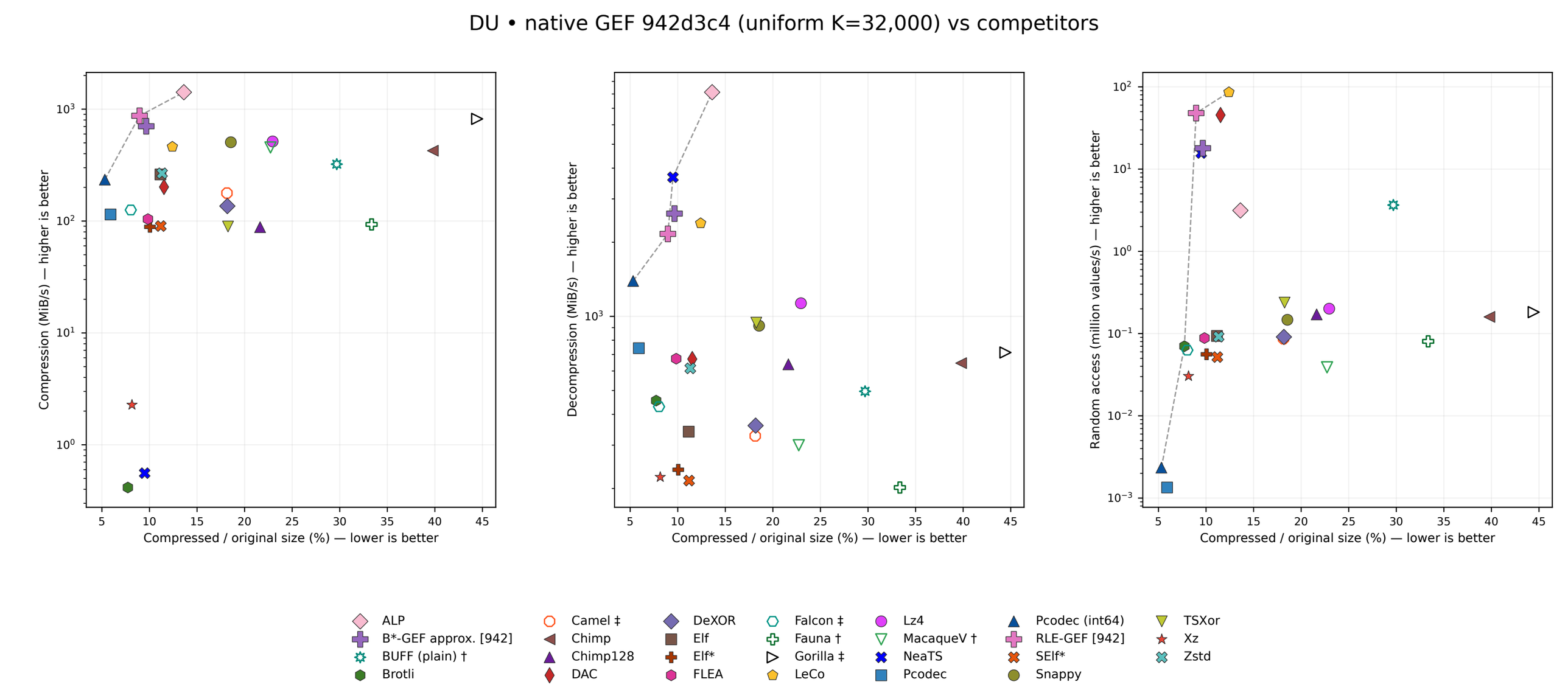
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

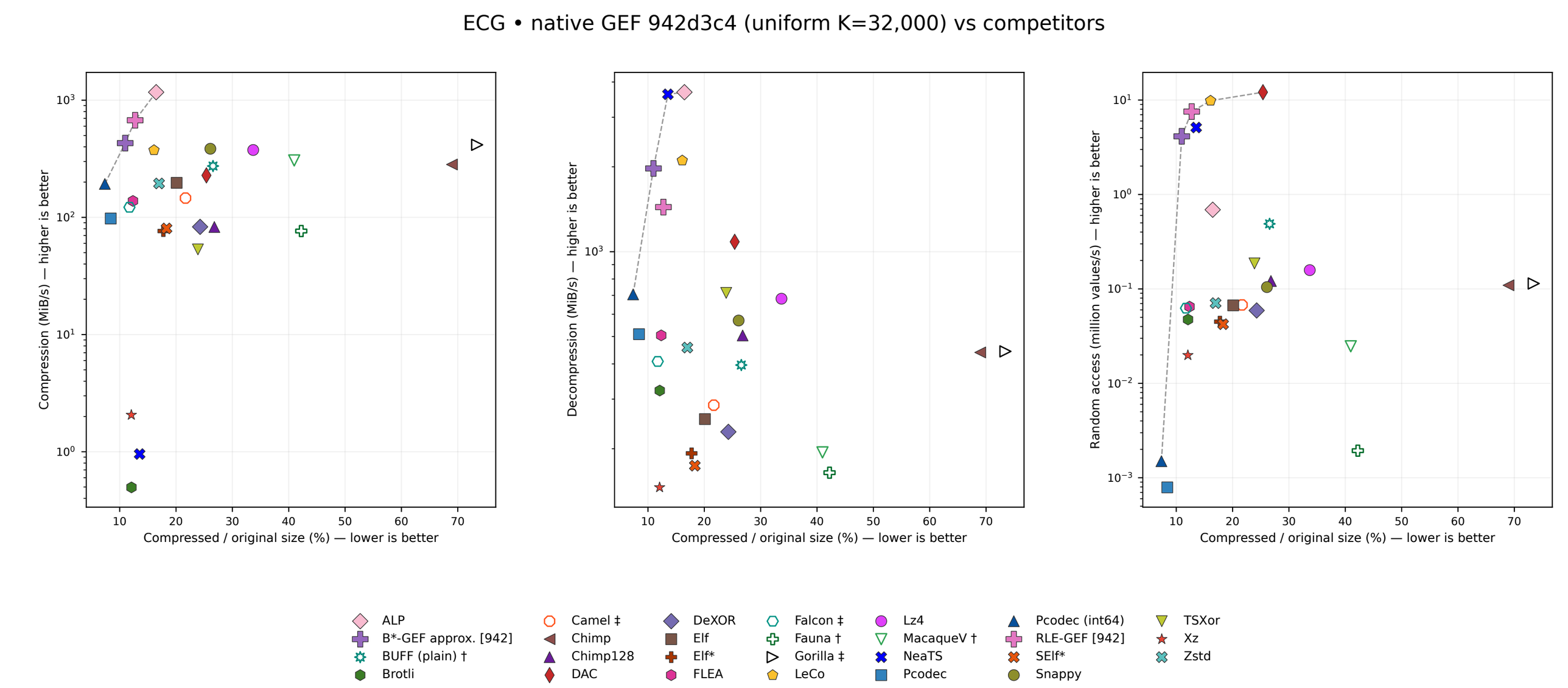
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

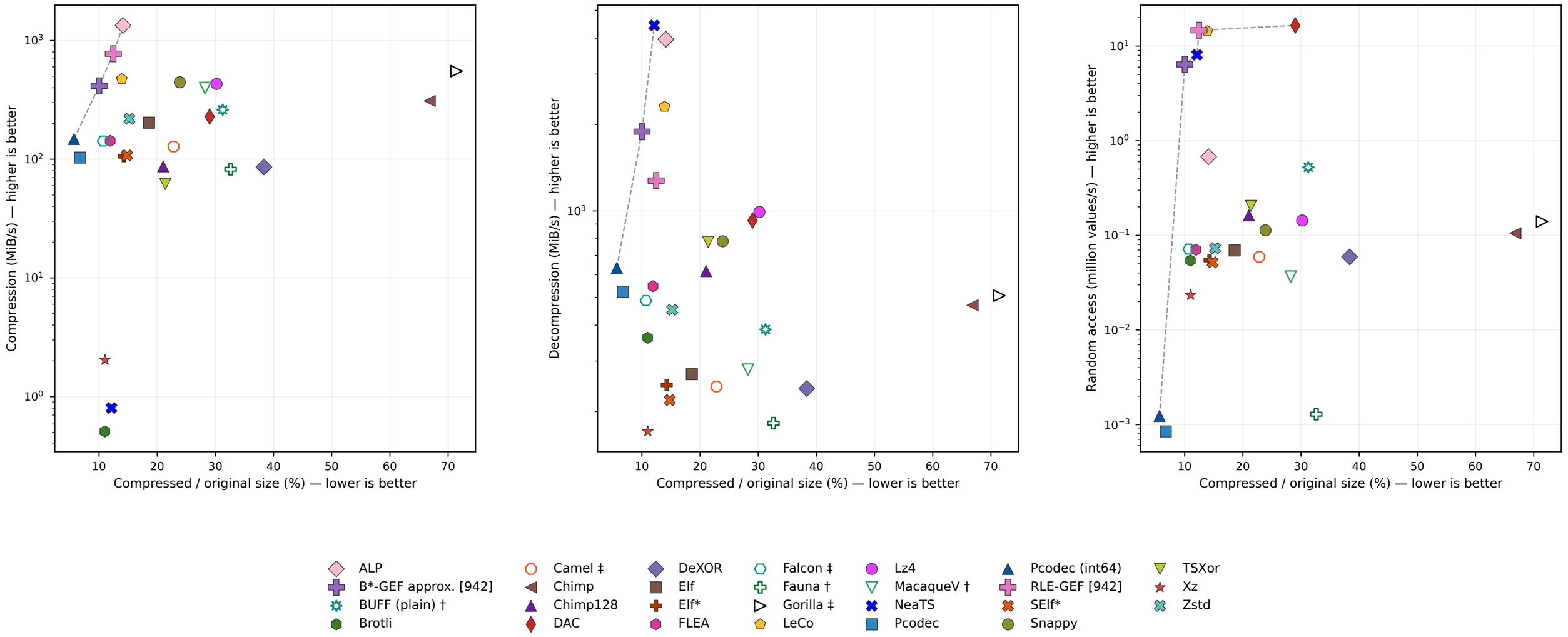
DP • native GEF 942d3c4 (uniform K=32,000) vs competitors



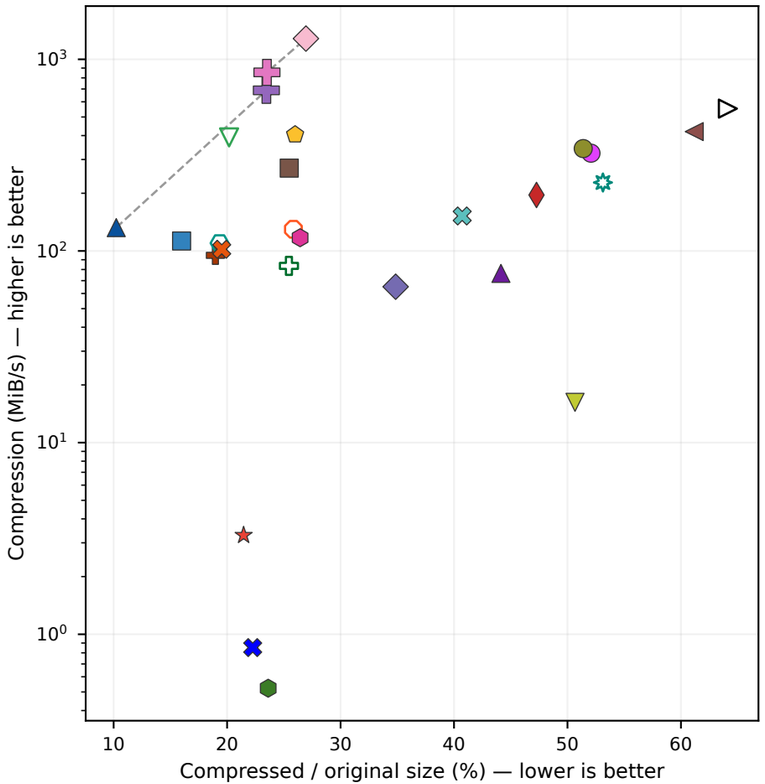
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

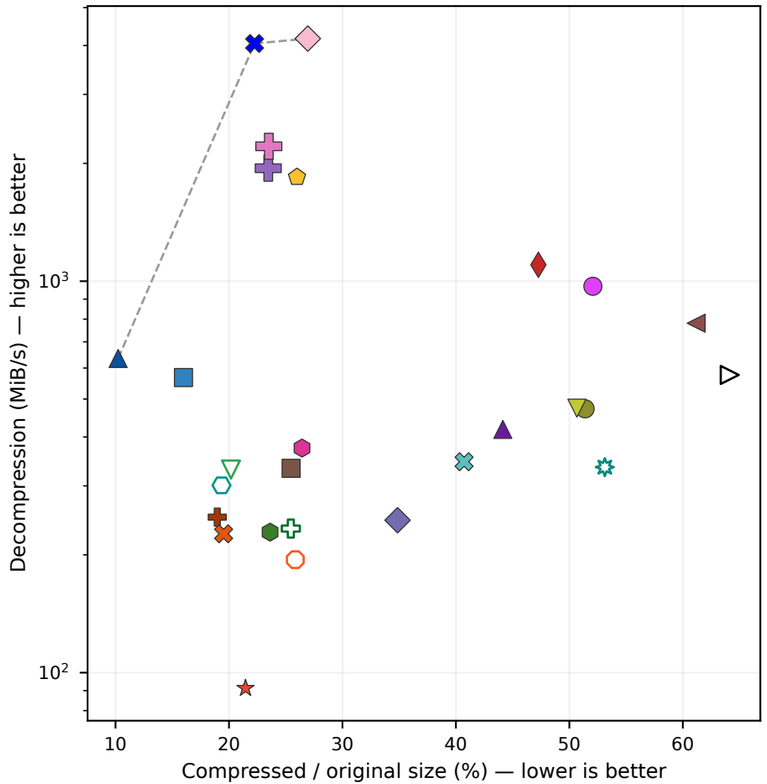

DU • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

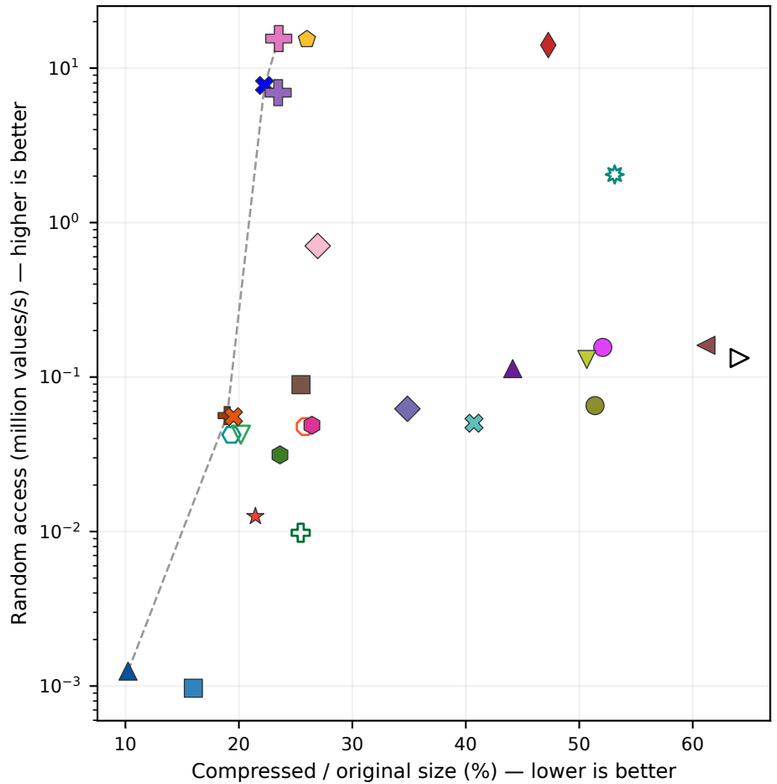

ECG • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.


GE • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

LAT • native GEF 942d3c4 (uniform K=32,000) vs competitors



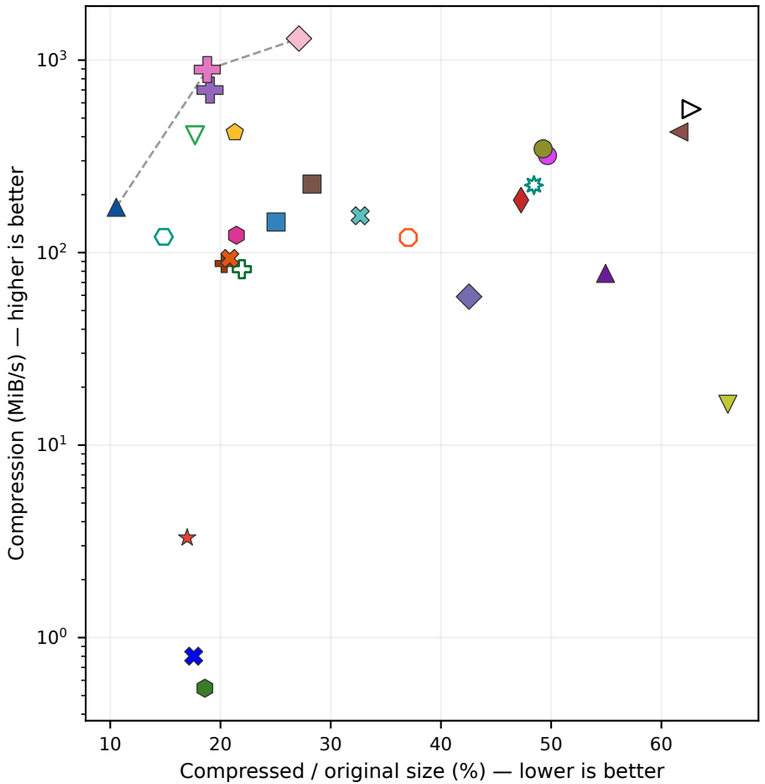


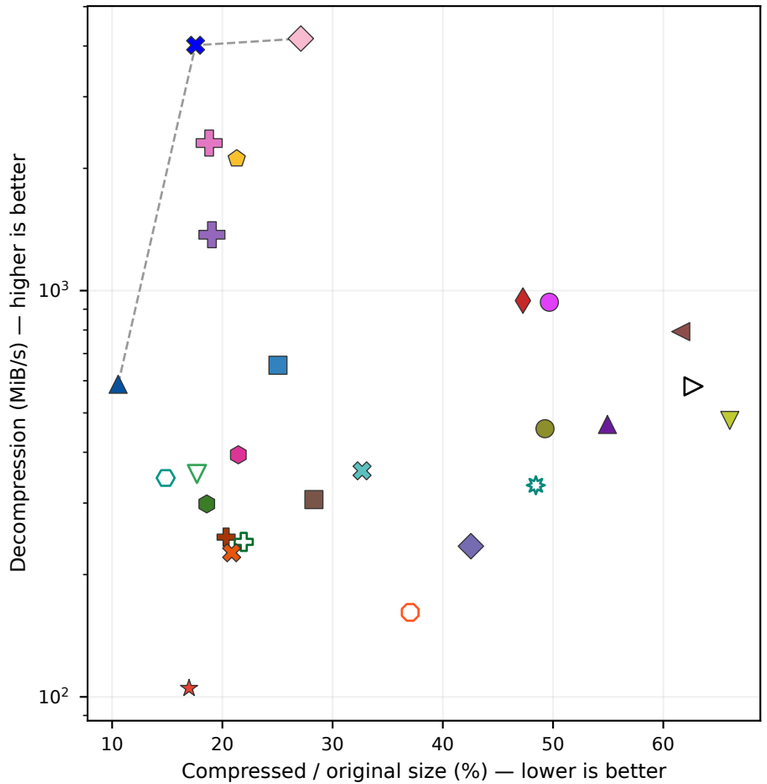


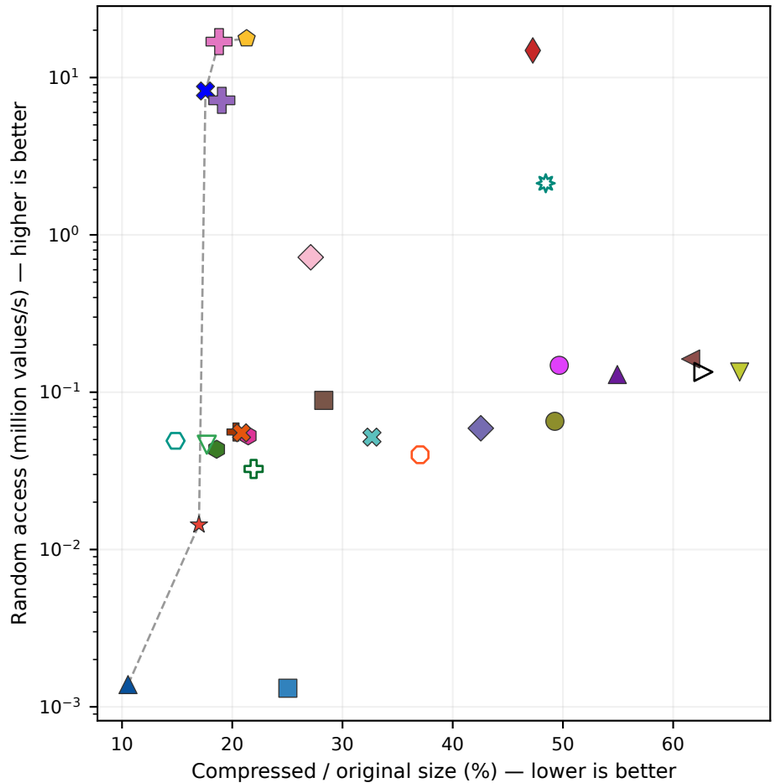
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

LON • native GEF 942d3c4 (uniform K=32,000) vs competitors

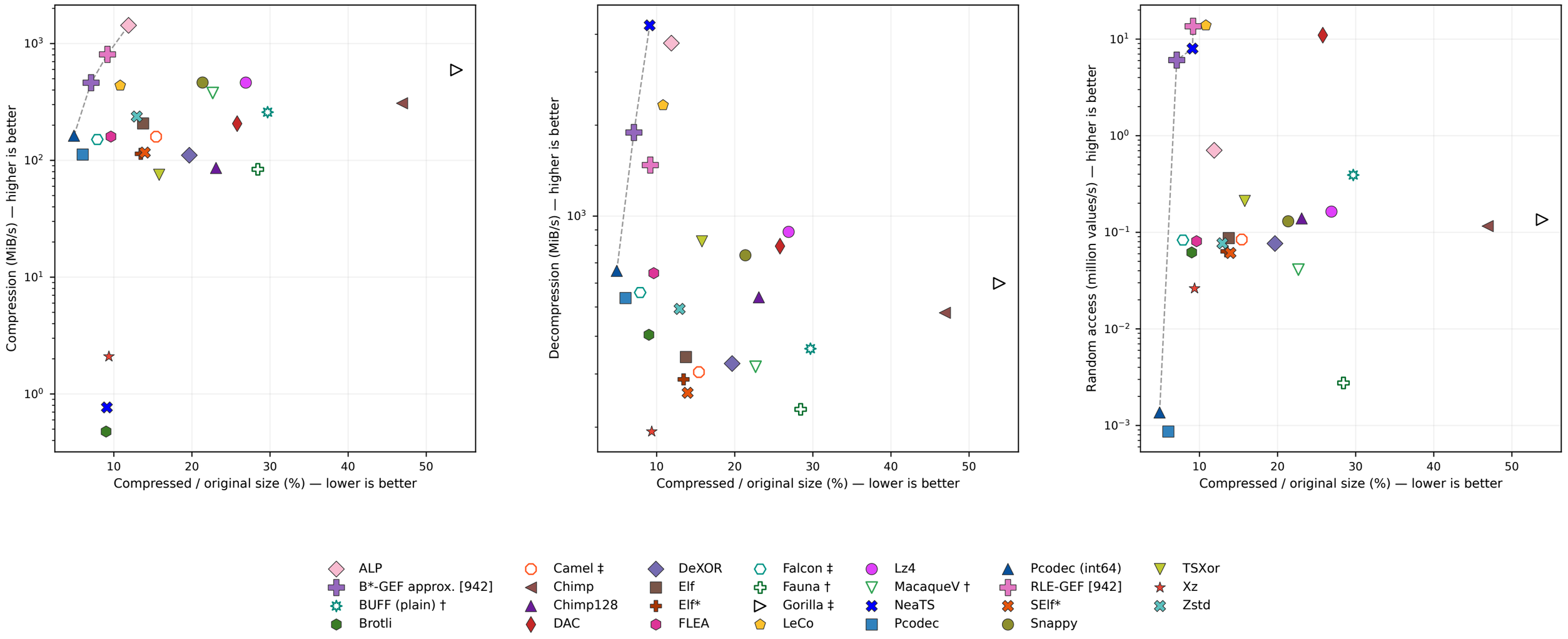




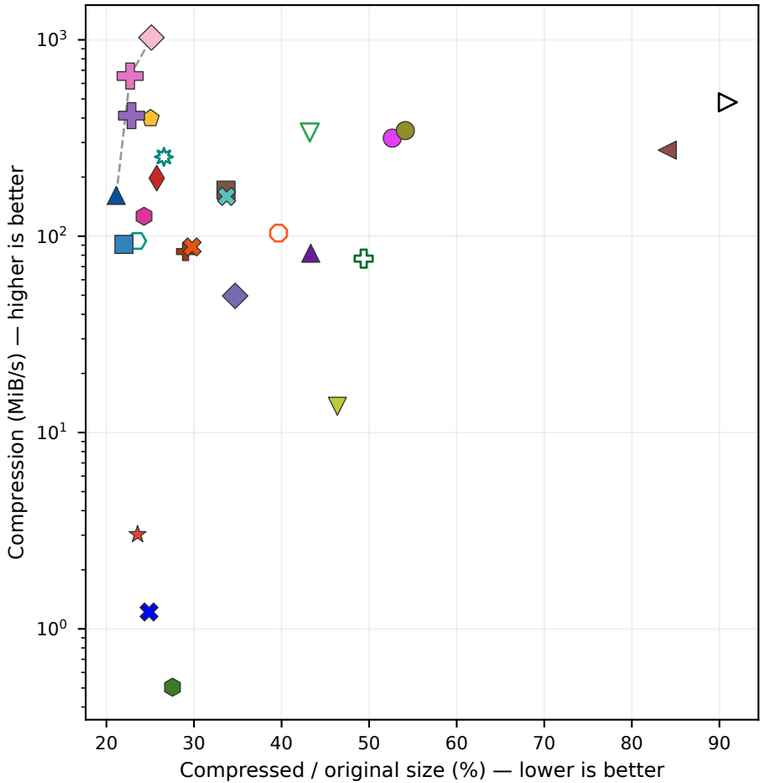


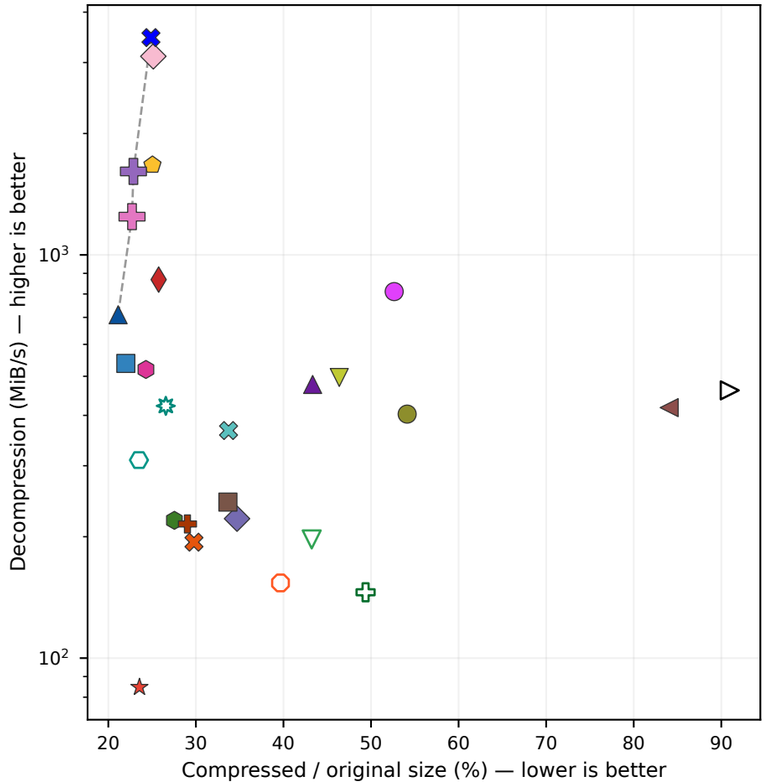
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

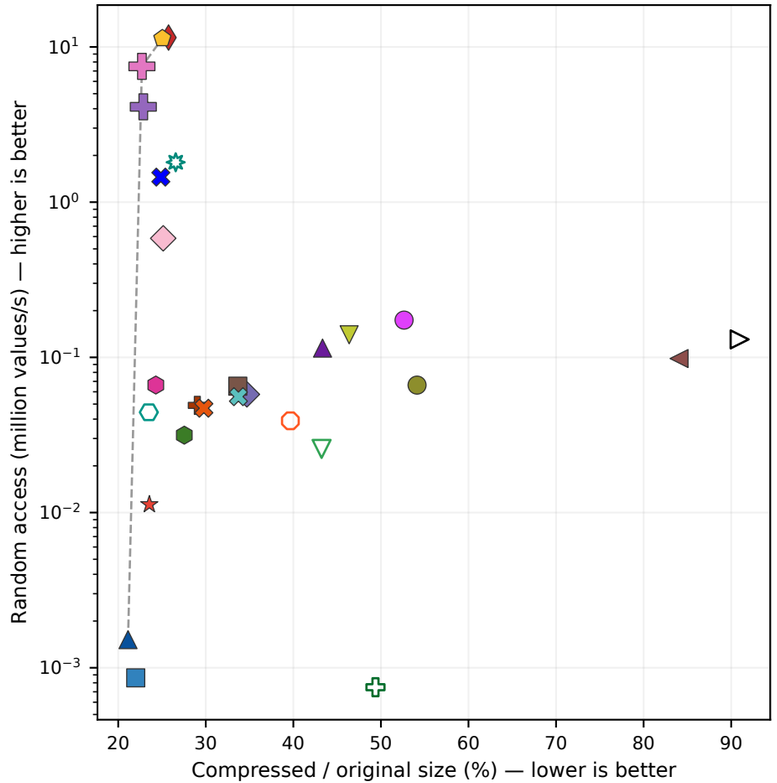
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.


UK • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.

WD • native GEF 942d3c4 (uniform K=32,000) vs competitors







ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Falcon ‡
Fauna †
Gorilla ‡
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP rerun 2026-11-05; other measurements retained.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Gorilla/Camel/Falcon: failed bitwise audit. Hollow points excluded.